# A01 Level: Foundation
structure → name

**Molecular Formula:** $C_6H_{14}$

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 5 C → pentane.

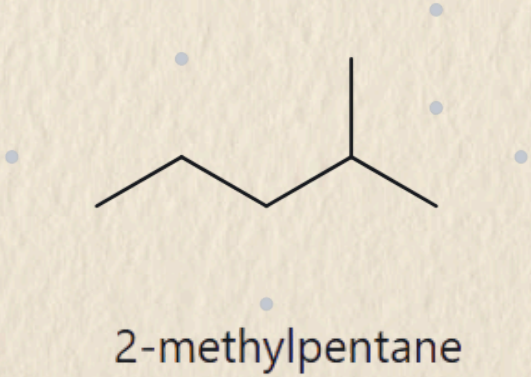
**Numbering:** left → right gives 4
right → left gives 2
First point of difference: 2 < 4, so number from the right.

**Substituents:**
methyl on C2 → 2-methyl

## 3. MY NAME

**2-methylpentane**

## 4. ALKANE CHECK




☑ match

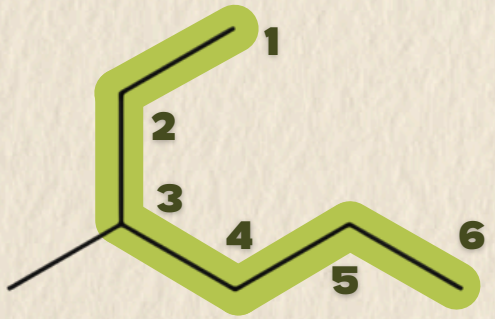
☑ longest chain ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# A02

**Level: Foundation**
structure → name

**Molecular Formula:** $C_7H_{16}$

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 6 C → hexane
It runs up through the ethyl arm, which is not the horizontal path (that one is only 5 C).

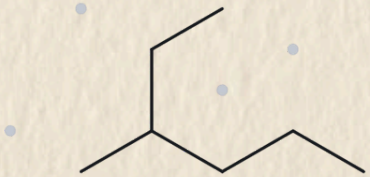
**Numbering:** ethyl end → other end gives 3
other end → ethyl end gives 4
First point of difference: 3 < 4, so number from the ethyl end.

**Substituents:**
methyl on C3 → 3-methyl

## 3. MY NAME

**3-methylhexane**

## 4. ALKANE CHECK



3-methylhexane

☑ **match**

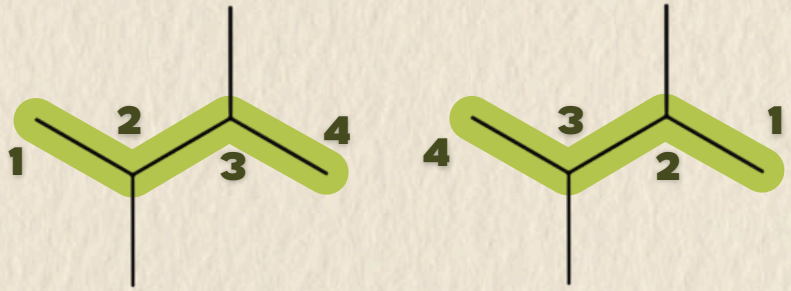
☑ longest chain ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# A03 Level: Foundation

structure → name

**Molecular Formula:** $C_6H_{14}$

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 4 C → butane

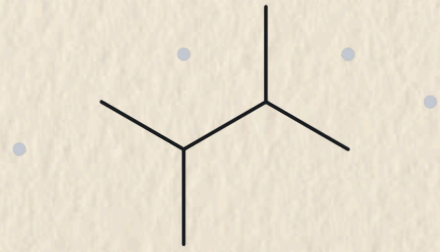
**Numbering:** left → right gives 2,3
right → left gives 2,3
There is no point of difference, so the direction does not matter.

**Substituents:**
methyl on C2 and C3 → 2,3-dimethyl

## 3. MY NAME

**2,3-dimethylbutane**

## 4. ALKANE CHECK



2,3-dimethylbutane

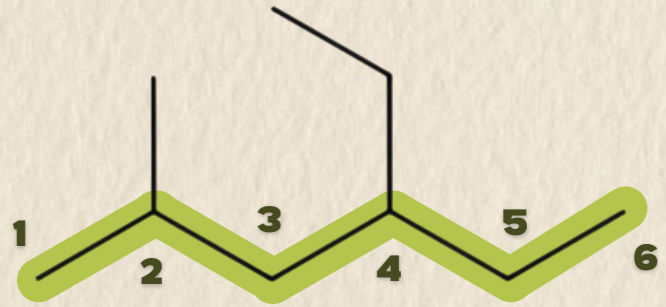
☑ match

☑ longest chain ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# A04 Level: Developing
structure → name

**Molecular Formula:** $C_9H_{20}$

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 6 C → hexane

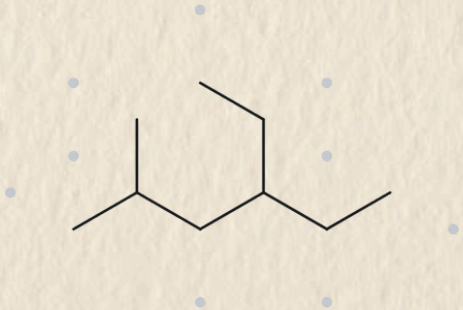
**Numbering:** left → right gives 2,4
right → left gives 3,5
First point of difference: 2 < 3, so number from the left.

**Substituents:**
ethyl on C4, methyl on C2.
Ethyl is cited first alphabetically → 4-ethyl-2-methyl

## 3. MY NAME

**4-ethyl-2-methylhexane**

## 4. ALKANE CHECK



4-ethyl-2-methylhexane

☑ **match**

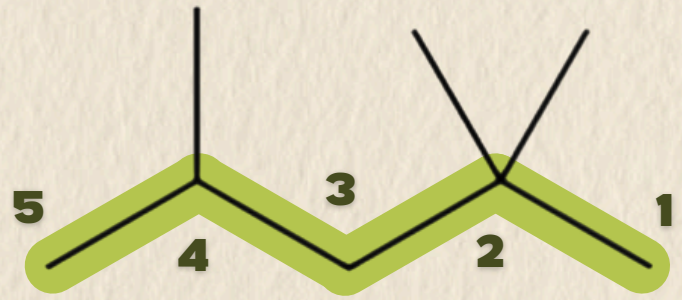
☑ longest chain ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# A05 Level: Developing

structure → name

**Molecular Formula:** $C_8H_{18}$

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 5 C → pentane

**Numbering:** left → right gives 2,4,4
right → left gives 2,2,4
The first locants tie (2 and 2). The point of difference is the second locant: 2 < 4, so number from the right.

**Substituents:**
methyl on C2, C2 and C4 → 2,2,4-trimethyl

## 3. MY NAME

**2,2,4-trimethylpentane**

## 4. ALKANE CHECK



2,2,4-trimethylpentane

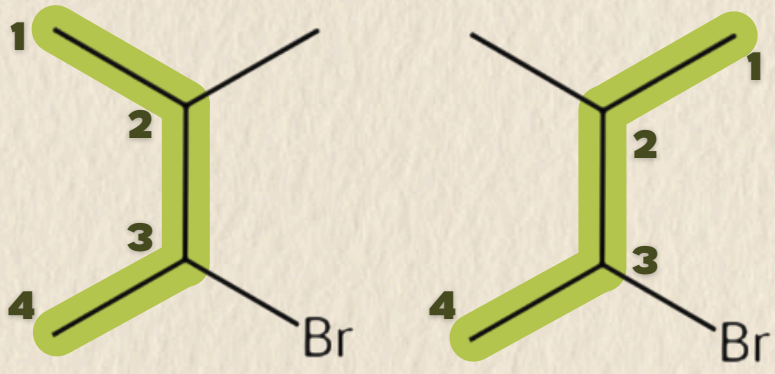
☑ match

☑ longest chain ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# A06 Level: Developing
structure → name

**Molecular Formula:** $C_5H_{11}Br$

## 1. NUMBERED PARENT




and two ways more

## 2. NAMING TRAIL

**Parent:** longest chain = 4 C → butane

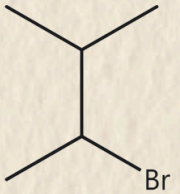
**Numbering:**
Both directions give locants 2 and 3.
Since the locant sets tie, use alphabetical order
Bromo comes before methyl, so bromo receives C2.

**Substituents:**
bromo on C2, methyl on C3 → 2-bromo-3-methyl

## 3. MY NAME

**2-bromo-3-methylbutane**

## 4. ALKANE CHECK




2-bromo-3-methylbutane

☑ **match**

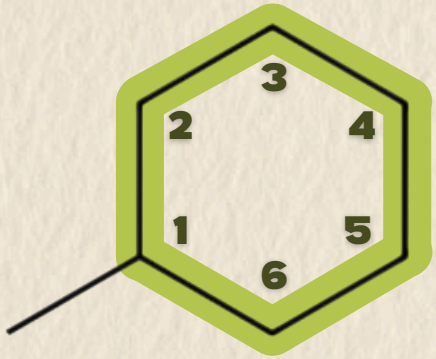
☑ longest chain ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# A07

**Level: Developing**
structure → name

**Molecular Formula:** $C_7H_{14}$

## 1. NUMBERED PARENT




The numbering is used only to count the parent ring carbons and identify the parent structure. It does not indicate a locant that needs to appear in the final name.

## 2. NAMING TRAIL

**Parent:** the ring is the parent, 6 C → cyclohexane

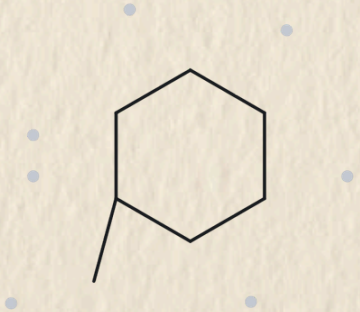
**Numbering:** only one substituent, so there is no point of difference and no locant is needed

**Substituents:** methyl → methyl

## 3. MY NAME

**methylcyclohexane**

## 4. ALKANE CHECK



methylcyclohexane

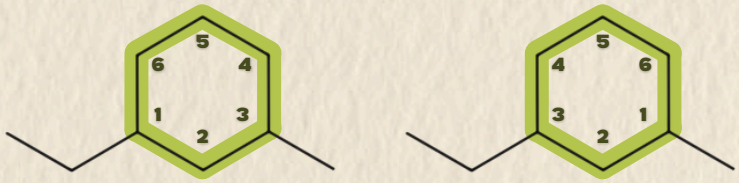
☑ **match.** The methyl bond angle differs, but the connectivity is the same (6-C ring, one methyl).

☑ ring confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# A08 Level: Challenge
structure → name

**Molecular Formula:** $C_9H_{18}$

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** the ring is the parent, 6 C → cyclohexane

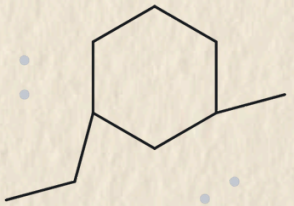
**Numbering:**
start at a substituent and go the shorter way round.
Ethyl C1 → methyl gives {1,3}; methyl C1 → ethyl gives {1,3}.
No point of difference, so the tie goes to alphabetical order: ethyl takes C1.

**Substituents:**
ethyl on C1, methyl on C3 → 1-ethyl-3-methyl

## 3. MY NAME

**1-ethyl-3-methylcyclohexane**

## 4. ALKANE CHECK



1-ethyl-3-methylcyclohexane

☑ **match.** The bond angles and lengths differ, but the connectivity is the same: a 6-C ring, ethyl and methyl on carbons 1,3 apart.

☑ ring confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# A09 Level: Challenge
structure → name

**Molecular Formula:** $C_{10}H_{22}$

## 1. NUMBERED PARENT




## 2. NAMING TRAIL

**Parent:** longest chain = 7 C → heptane
(the path through the branch is only 6 C)

**Numbering:** left → right gives 4; right → left gives 4. There is no point of difference, so the direction does not matter.

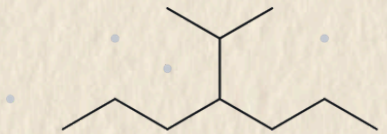
**Substituents:**
one 3-carbon branch attached through its middle carbon (isopropyl) on C4 → 4-(1-methylethyl)

## 3. MY NAME

**4-(1-methylethyl)heptane**

also acceptable: 4-isopropylheptane or 4-(propan-2-yl)heptane

## 4. ALKANE CHECK



2-methyl-3-propylhexane

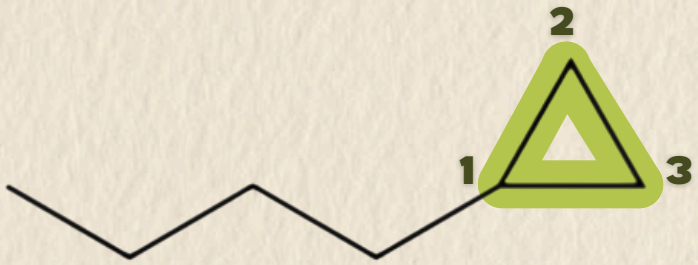
☒ **mismatch.** Correct name: 4-(1-methylethyl)heptane. The parent is the longest continuous chain. The tool chose a 6-C chain (hexane), but a 7-C chain (heptane) exists.

☑ longest chain ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# A10 Level: Challenge
structure → name

**Molecular Formula:** $C_7H_{14}$

## 1. NUMBERED PARENT




The numbering is used only to count the parent ring carbons and identify the parent structure. It does not indicate a locant that needs to appear in the final name.

## 2. NAMING TRAIL

**Parent:** the ring (3C) is the parent (2013 rule), even though the chain is longer → cyclopropane

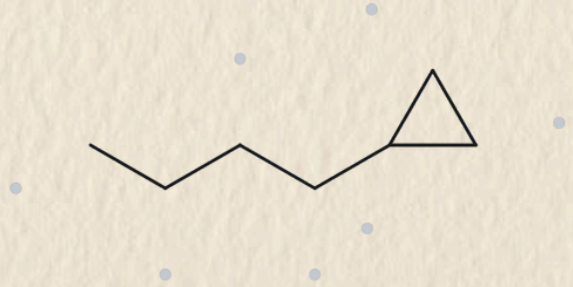
**Numbering:** only one substituent, so there is no point of difference and no locant is needed

**Substituents:**
4-carbon chain → butyl → butyl

## 3. MY NAME

**butylcyclopropane**

## 4. ALKANE CHECK



butylcyclopropane

☑ match

☑ ring confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# B01 Level: Foundation
structure → name

GIVEN NAME:
# 2-methylpropane

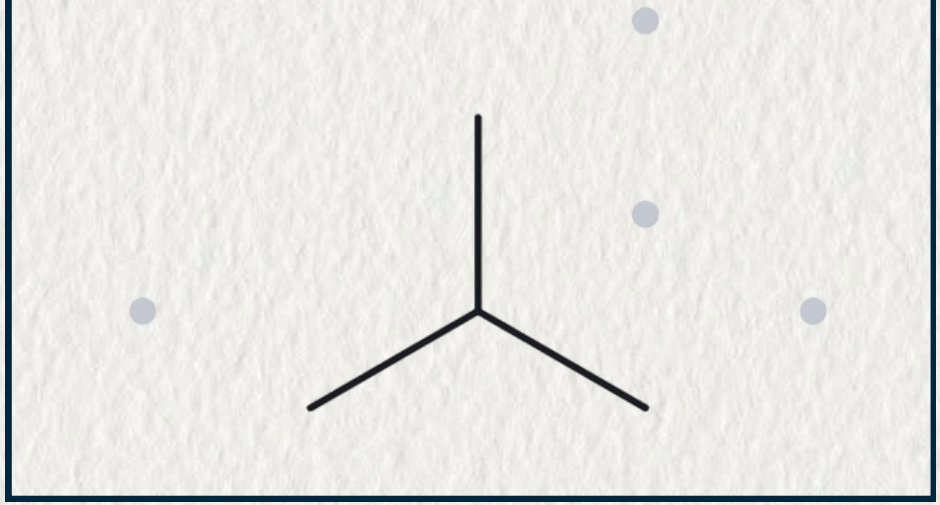
## 1. NAME DECODED

**Parent:** propane → 3-C chain

**Substituents:** methyl on C2

**Expected Formula (using the general formula):**
Total carbon atoms = 3 + 1 = 4
***open chain:*** $C_nH_{2n+2} = C_4H_{2(4)+2} = \mathbf{C_4H_{10}}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)



## 3. FORMULA CHECK

**Counted from the drawing:**

$C_4H_{10}$

**General Formula:**

$C_4H_{10}$

**counted = general formula**

## 4. BACK-CHECK

**Returned by The Alkane:**

2-methylpropane

☑ **same as the given name**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# B02 Level: Foundation
structure → name

GIVEN NAME:
# 3-ethylpentane

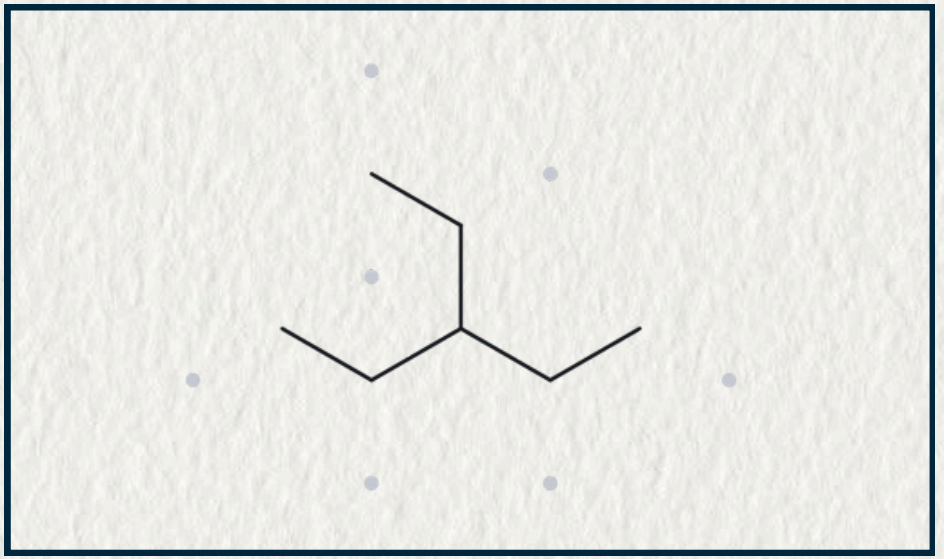
## 1. NAME DECODED

**Parent:** pentane → 5-C chain

**Substituents:** ethyl on C3

**Expected Formula (using the general formula):**
Total carbon atoms = 5 + 2 = 7
***open chain:*** $C_nH_{2n+2}$ = $C_7H_{2(7)+2}$ = **$C_7H_{16}$**

## 2. BOND-LINE STRUCTURE (THE ALKANE)



## 3. FORMULA CHECK

**Counted from the drawing:**

$C_7H_{16}$

**General Formula:**

$C_7H_{16}$

**counted = general formula**

## 4. BACK-CHECK

**Returned by The Alkane:**

3-ethylpentane

☑ **same as the given name**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# B03 Level: Foundation
structure → name

GIVEN NAME:
# 2,2-dimethylbutane

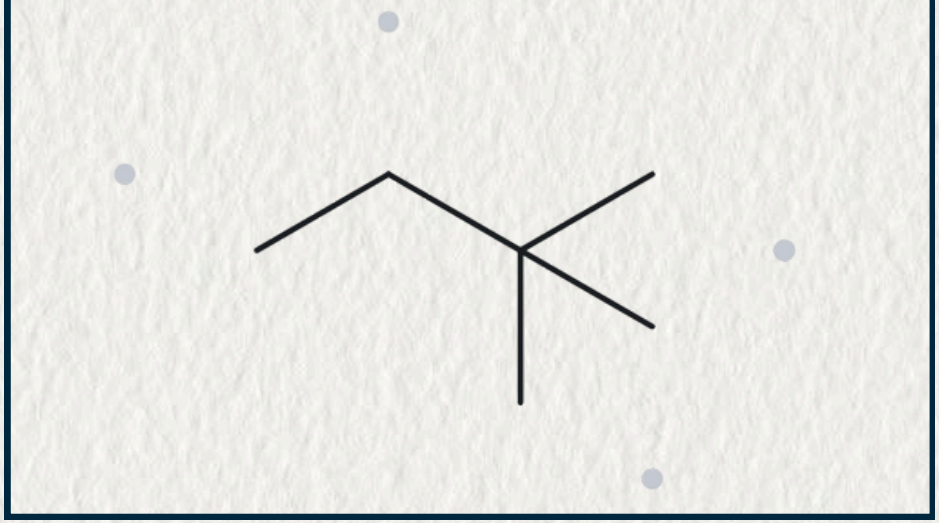
## 1. NAME DECODED

**Parent:** butane → 4-C chain

**Substituents:** two methyls on C2

**Expected Formula (using the general formula):**
Total carbon atoms = 4 + 2 = 6
***open chain:*** $C_nH_{2n+2}$ = $C_6H_{2(6)+2}$ = **$C_6H_{14}$**

## 2. BOND-LINE STRUCTURE (THE ALKANE)



## 3. FORMULA CHECK

**Counted from the drawing:**

$C_6H_{14}$

**General Formula:**

$C_6H_{14}$

**counted = general formula**

## 4. BACK-CHECK

**Returned by The Alkane:**

2,2-dimethylbutane

☑ **same as the given name**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# B04 Level: Developing
structure → name

GIVEN NAME:
# 3-ethyl-2-methylpentane

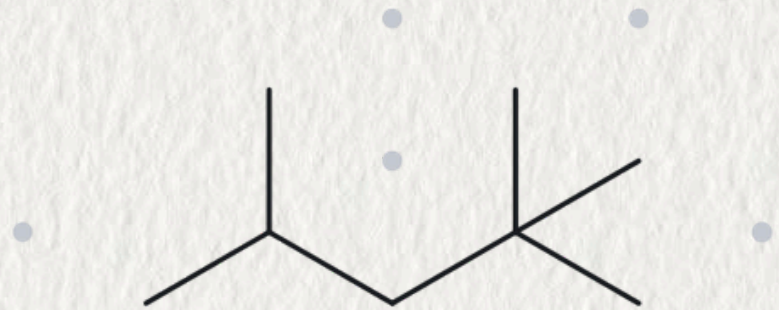
## 1. NAME DECODED

**Parent:** pentane → 5-C chain

**Substituents:** methyl on C2, ethyl on C3

**Expected Formula (using the general formula):**
Total carbon atoms = 5 + 1 + 2= 8
***open chain:*** $C_nH_{2n+2}$ = $C_8H_{2(8)+2}$ = **$C_8H_{18}$**

## 2. BOND-LINE STRUCTURE (THE ALKANE)



## 3. FORMULA CHECK

**Counted from the drawing:**

$C_8H_{18}$

**General Formula:**

$C_8H_{18}$

**counted = general formula**

## 4. BACK-CHECK

**Returned by The Alkane:**

3-ethyl-2-methylpentane

☑ **same as the given name**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# B05 Level: Developing
structure → name

GIVEN NAME:
# 1-chloro-2-methylpropane

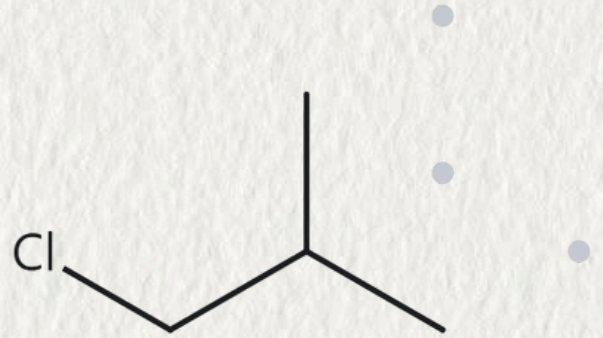
## 1. NAME DECODED

**Parent:** propane → 3-C chain

**Substituents:** chloro on C1, methyl on C2

**Expected Formula (using the general formula):**
Total carbon atoms = 3 + 1 = 4
***open chain:*** $C_nH_{2n+2} = C_4H_{2(4)+2} = C_4H_{10}$
replaces one H → **$C_4H_9Cl$**

## 2. BOND-LINE STRUCTURE (THE ALKANE)




## 3. FORMULA CHECK

**Counted from the drawing:**

$C_4H_9Cl$

**General Formula:**

$C_4H_9Cl$

**counted = general formula**

## 4. BACK-CHECK

**Returned by The Alkane:**

1-chloro-2-methylpropane

☑ **same as the given name**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# B06 Level: Developing
structure ➜ name

GIVEN NAME:
# 1,2-dimethylcyclopentane

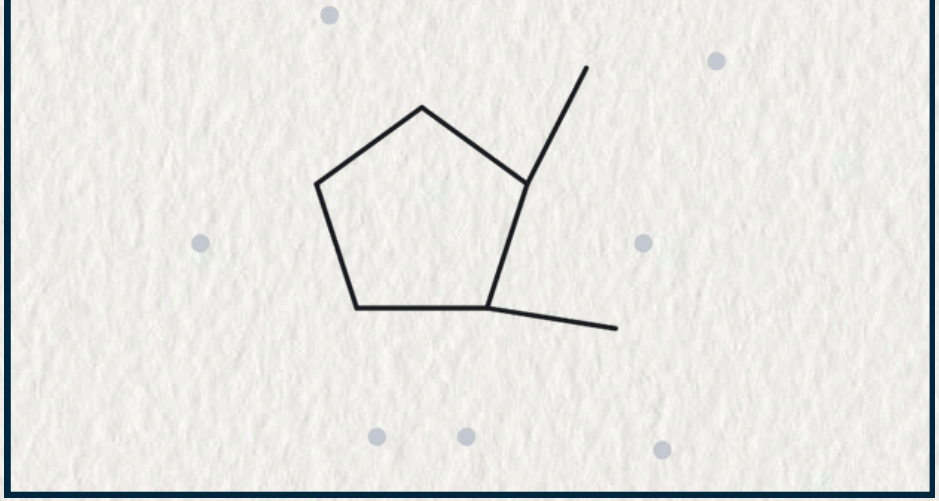
## 1. NAME DECODED

**Parent:** cyclopentane → 5-C ring

**Substituents:**
methyl on C1 and C2 (adjacent carbons)

**Expected Formula (using the general formula):**
Total carbon atoms = 5 + 2 = 7
***one ring:*** $C_nH_{2n}$ = $C_4H_{2(4)}$ = **$C_7H_{14}$**

## 2. BOND-LINE STRUCTURE (THE ALKANE)



## 3. FORMULA CHECK

**Counted from the drawing:**

$C_7H_{14}$

**General Formula:**

$C_7H_{14}$

**counted = general formula**

## 4. BACK-CHECK

**Returned by The Alkane:**

1,2-dimethylcyclopentane

☑ **same as the given name**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

**B07** Level: Developing
structure ➜ name

GIVEN NAME:

# 4-ethyl-3,3-dimethylheptane

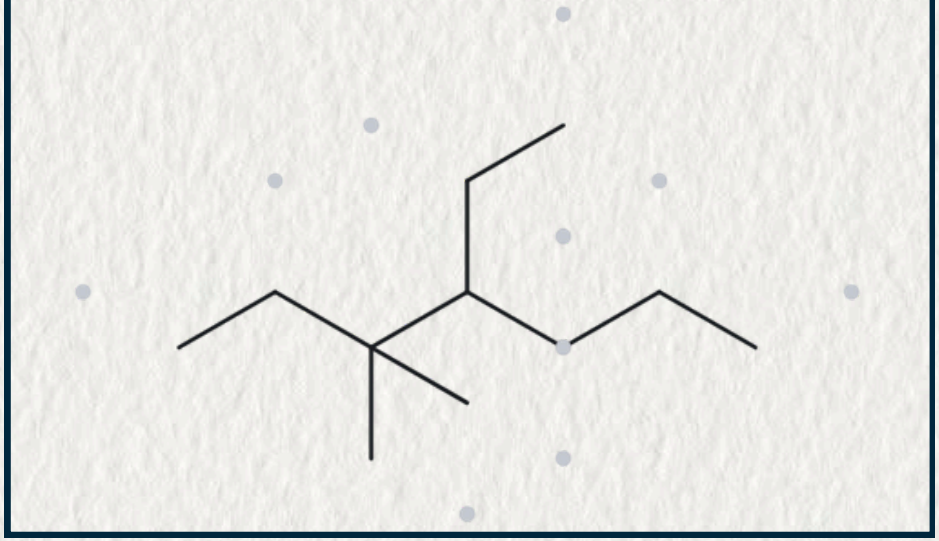
## 1. NAME DECODED

**Parent:** heptane → 7-C chain

**Substituents:** two methyls on C3, ethyl on C4

**Expected Formula (using the general formula):**
Total carbon atoms = 7 + 2 + 2 = 11
***open chain:*** $C_nH_{2n+2} = C_{11}H_{2(11)+2} = \mathbf{C_{11}H_{24}}$

## 2. BOND-LINE STRUCTURE (THE ALKANE)



## 3. FORMULA CHECK

**Counted from the drawing:**

$C_{11}H_{24}$

**General Formula:**

$C_{11}H_{24}$

**counted = general formula**

## 4. BACK-CHECK

**Returned by The Alkane:**

4-ethyl-3,3-dimethylheptane

☑ **same as the given name**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# B08 Level: Challenge
structure ➜ name

GIVEN NAME:
# tert-butylcyclohexane

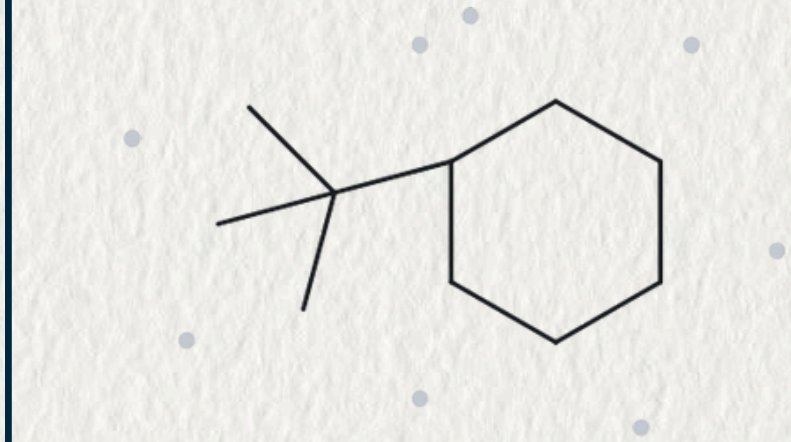
## 1. NAME DECODED

**Parent:** cyclohexane → 6-C ring

**Substituents:**
tert-butyl, a carbon bearing three methyls,
$-C(CH_3)_3$ → 4 C

**Expected Formula (using the general formula):**
Total carbon atoms = 6 + 4 = 10
***one ring:*** $C_nH_{2n}$ = $C_{10}H_{2(10)}$ = **$C_{10}H_{20}$**

## 2. BOND-LINE STRUCTURE (THE ALKANE)



## 3. FORMULA CHECK

**Counted from the drawing:**

$C_{10}H_{20}$

**General Formula:**

$C_{10}H_{20}$

**counted = general formula**

## 4. BACK-CHECK

**Returned by The Alkane:**

(1,1-dimethylethyl)cyclohexane

☑ **Same compound as the given name. The Alkane writes tert-butyl as (1,1-dimethylethyl), the bracketed form of tert-butylcyclohexane.**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# B09 Level: Challenge
structure ➜ name

GIVEN NAME:
# 3-ethyl-1,1-dimethylcyclohexane

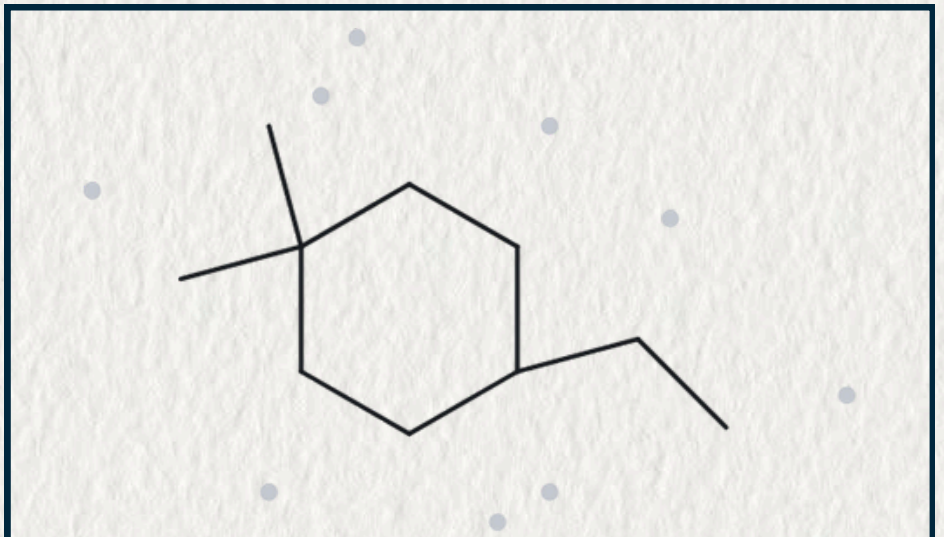
## 1. NAME DECODED

**Parent:** cyclohexane → 6-C ring

**Substituents:** two methyls on C1, ethyl on C3

**Expected Formula (using the general formula):**
Total carbon atoms = 6 + 2 + 2 = 10
***one ring:*** $C_nH_{2n}$ = $C_{10}H_{2(10)}$ = **$C_{10}H_{20}$**

## 2. BOND-LINE STRUCTURE (THE ALKANE)



## 3. FORMULA CHECK

**Counted from the drawing:**

$C_{10}H_{20}$

**General Formula:**

$C_{10}H_{20}$

**counted = general formula**

## 4. BACK-CHECK

**Returned by The Alkane:**

3-ethyl-1,1-dimethylcyclohexane

☑ **same as the given name**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# B10 Level: Challenge
structure ➜ name

GIVEN NAME:
## 2-ethylbutane
deliberately incorrect name

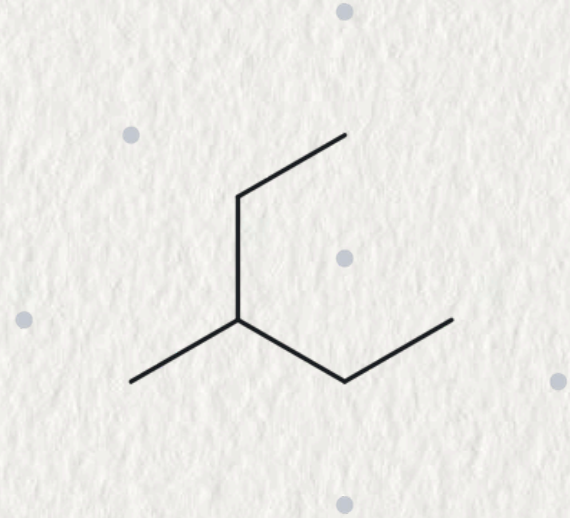
### 1. NAME DECODED

**Parent:** butane → 4-C chain

**Substituents:** ethyl on C2

**Expected Formula (using the general formula):**
Total carbon atoms = 4 + 2 = 6
***open chain:*** $C_nH_{2n+2} = C_6H_{2(6)+2} = \mathbf{C_6H_{14}}$

### 2. BOND-LINE STRUCTURE (THE ALKANE)



### 3. FORMULA CHECK

**Counted from the drawing:**

$C_6H_{14}$

**General Formula:**

$C_6H_{14}$

**counted = general formula**

### 4. BACK-CHECK

**Returned by The Alkane:**

2-methylpropane

☒ **Not the same as the given name.** **The parent is the longest continuous chain. Through the ethyl group, there is a 5-C chain, so the parent is pentane with a methyl on C3.**

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded